

Adaptive Activation Steering for Efficient LLM Reasoning via Closed-Loop PID Control

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Abstract

Reasoning LLMs trained with long chain-of-thought often overthink: they spend tokens on redundant reflection and transitions that inflate cost without improving accuracy. Static activation steering (e.g. SEAL) suppresses such content with a fixed vector, but applies the same strength regardless of how redundant the current chunk actually is. We describe *PID-steering*, a training-free, decoding-time method that modulates the steering strength with a PID controller driven by a lightweight chunk-level redundancy classifier. On a 77-problem GSM8K subset with DeepSeek-R1-Distill-Qwen-1.5B, the method improves accuracy from 85.7% to 89.6% (+3.9 pp) while cutting average output length from 1026 to 790 tokens (−23%). We report it as a small-scale proof of concept rather than a benchmark result.

1 Introduction

Long chain-of-thought reasoning [1] underlies the gains of o1-style and DeepSeek-R1-style models [2], but the same training signal encourages overthinking: models emit long runs of reflection and transition text that add cost and sometimes hurt accuracy [3]. Most remedies—length-reward RL, variable-length SFT, latent-CoT compression—require retraining the model.

SEAL [4] instead steers activations at inference time with a fixed vector built from the difference between “execution” and “reflection/transition” chunks. This is cheap and training-free, but the steering coefficient is constant: the same push is applied to a chunk that is clearly redundant and to one that is doing useful work. A constant coefficient strong enough to suppress the worst chunks can therefore over-suppress productive reasoning, while a gentler one leaves redundancy untouched.

We close the loop. A chunk-level classifier estimates the redundancy probability of the current reasoning chunk, and a PID controller uses the error between that probability and a target to set the steering coefficient on the fly. The method is training-free beyond fitting a small linear classifier and tuning four gains on a validation set.

Contributions.

- A closed-loop, decoding-time steering scheme that treats activation steering as a control problem rather than a fixed intervention.
- A chunk-level redundancy classifier, trained on a small set of labeled math-reasoning chunks, that serves as the PID feedback signal.
- On GSM8K (77 problems, DeepSeek-R1-Distill-Qwen-1.5B): +3.9 pp accuracy (85.7 → 89.6%) and −23% tokens (1026 → 790) over the unsteered baseline.

2 Related Work

Efficient reasoning. Sui et al. [3] survey the space: length-reward RL, variable-length SFT, latent-CoT compression, and prompt-level controls. These act at training time or rewrite the reasoning format; our method acts at decoding time on a frozen model.

Activation steering. SEAL [4] categorizes reasoning chunks as execution, reflection, or transition, and subtracts a fixed steering vector to suppress the latter two. We reuse SEAL’s vector construction but replace the fixed coefficient with a feedback controller. This is what lets the method improve accuracy and cut tokens together, rather than trading one against the other.

3 Method

3.1 Setup

At decoding step t , let $h_t \in \mathbb{R}^d$ be the hidden state at a chosen layer l . Steering replaces h_t with

$$h'_t = h_t + \alpha_t \cdot v,$$

where v is a fixed steering direction and $\alpha_t \geq 0$ is a time-varying coefficient chosen by the controller. Following SEAL [4],

$$v = \mathbb{E}[h_l \mid \text{redundant chunk}] - \mathbb{E}[h_l \mid \text{required chunk}],$$

estimated from a small labeled set.

3.2 Chunk-level redundancy classifier

We split the generation into fixed-size chunks of 24 tokens and mean-pool the layer- l hidden states over each chunk. A logistic classifier C outputs

$$p_{\text{red},t} = C(\bar{h}_l^{\text{chunk}(t)}) \in [0, 1].$$

The classifier is the only learned component, and its cost is negligible next to a decoder step.

3.3 PID controller

Let $e_t = p_{\text{red},t} - p_{\text{target}}$ be the redundancy error. The controller maintains

$$\begin{aligned} I_t &= \text{clip}(I_{t-1} + K_I e_t, -I_{\max}, I_{\max}), \\ D_t &= K_D(e_t - e_{t-1}), \\ \alpha_t &= \text{clip}(\alpha_{t-1} + K_P e_t + I_t + D_t, 0, \alpha_{\max}). \end{aligned}$$

The integral term is clipped to prevent windup, and α_t is floored at 0 so the controller can only *push away* from redundancy, never amplify it. An ϵ -margin on e_t avoids reacting to classifier noise near the target. We also couple the sampling temperature to the steering strength, interpolating linearly from 0.6 at $\alpha_t = 0$ down to 0.3 at $\alpha_t = \alpha_{\max}$, so that stronger suppression is paired with more deterministic decoding.

4 Experiments

Model and data. DeepSeek-R1-Distill-Qwen-1.5B, evaluated on a 77-problem subset of the GSM8K test set.

Classifier. Logistic regression (SGD, log loss) on mean-pooled layer-20 hidden states; chunk size 24 tokens (selected from $\{16, 24\}$); trained on labeled chunks from GSM8K training problems.

Algorithm 1 Closed-loop PID steering at decoding time

```
1:  $\alpha \leftarrow 0, I \leftarrow 0, e_{\text{prev}} \leftarrow 0$ 
2: for generation step  $t = 1, 2, \dots$  do
3:   if  $t \geq t_{\text{init}}$  and within steering window then
4:      $p_{\text{pred}} \leftarrow C(\bar{h}_l^{\text{chunk}(t)})$ 
5:      $e \leftarrow p_{\text{pred}} - p_{\text{target}}$ 
6:     if  $e > \epsilon_{\text{margin}}$  then
7:        $I \leftarrow \text{clip}(I + K_I e, -I_{\text{max}}, I_{\text{max}})$ 
8:        $D \leftarrow K_D(e - e_{\text{prev}})$ 
9:        $\alpha \leftarrow \text{clip}(\alpha + K_P e + I + D, 0, \alpha_{\text{max}})$ 
10:    end if
11:     $e_{\text{prev}} \leftarrow e$ 
12:  end if
13:   $h'_t \leftarrow h_t + \alpha \cdot v$ ; sample next token from  $h'_t$ 
14: end for
```

Hyperparameters. $K_P = 0.05$, $K_I = 0.001$, $K_D = 0.001$, $p_{\text{target}} = 0.5$, $\alpha_{\text{max}} = 0.4$, $I_{\text{max}} = 0.2$, $\epsilon_{\text{margin}} = 0.2$, free period $t_{\text{init}} = 80$ tokens, steering window 60 tokens, sampling temperature 0.3–0.6 (coupled to α_t), maximum generation 4096 tokens.

Metrics. Exact-match accuracy against the GSM8K gold answer and mean generated tokens per problem.

4.1 Main result

Table 1: GSM8K, 77 problems, DeepSeek-R1-Distill-Qwen-1.5B.

Method	Accuracy (%)	Avg. tokens
Baseline (no steering)	85.7	1026
PID-steering	89.6	790

The method improves accuracy by 3.9 percentage points while using 23% fewer tokens (Table 1). Both metrics move in the same direction, which is the main empirical point: adaptive steering can suppress redundancy without the accuracy cost that a constant coefficient incurs once it is strong enough to be useful on the worst chunks.

5 Limitations

The evaluation covers 77 problems, one dataset, and one 1.5B model, so the numbers should be read as a proof of concept rather than a benchmark result. The classifier is trained on math chunks and is unlikely to transfer to non-math reasoning without new labels. The PID gains and the target p_{target} were tuned by hand on a small set; behavior under automated tuning or larger models is not yet characterized. We also do not ablate the PID terms individually, so the relative contribution of K_P , K_I , and K_D is unknown.

6 Conclusion

We frame activation steering as a closed-loop control problem: a small redundancy classifier supplies feedback, and a PID controller sets the steering coefficient per chunk. On a 77-problem GSM8K slice with DeepSeek-R1-Distill-Qwen-1.5B, this replaces a +accuracy/−tokens tradeoff with a joint

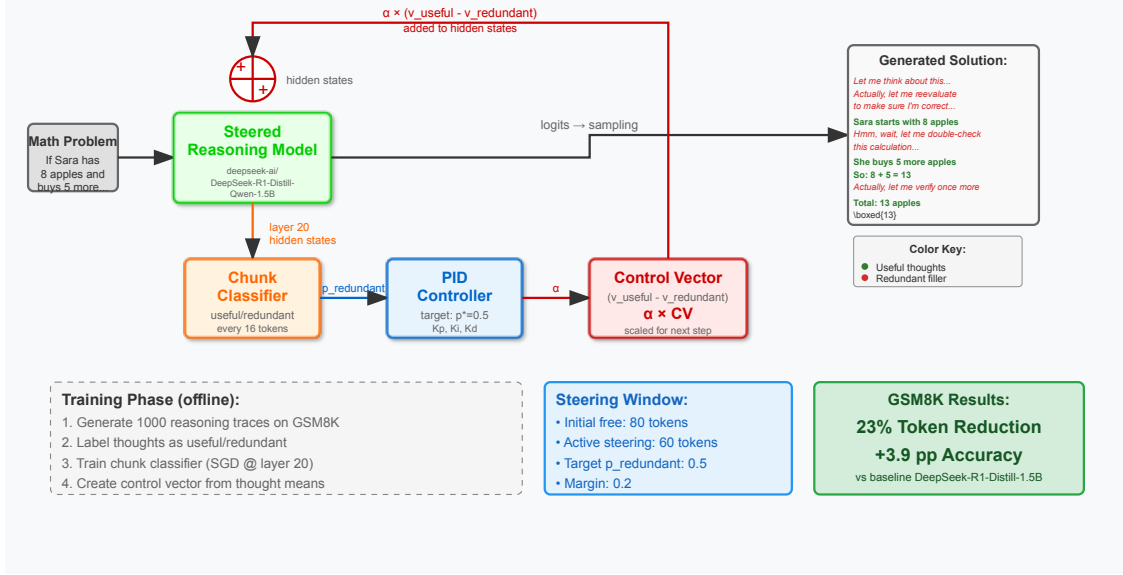


Figure 1: Method overview. The classifier scores each reasoning chunk; the PID controller turns the redundancy error into a steering coefficient α_t applied to the layer- l residual stream.

+accuracy/+token-savings improvement (85.7 $\rightarrow$ 89.6%, 1026 $\rightarrow$ 790 tokens). Whether the approach holds up at larger scale and on non-math reasoning is the obvious next question.

Code and results: https://github.com/arbwdj/pid_steering

References

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